"The institute, in print"

The imsi exchange

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Latest News

Academic Year 2026-2027

On the occasion of the 2026-2027 academic year, the IMSI Exchange wishes to extend its sincere greetings to all members of our community - students, faculty, technical and administrative staff, and representatives of scientific clubs. May this new academic year be marked by rigor, curiosity, and collaboration. The newspaper remains an open space for your contributions and feedback, so that it may continue to reflect the scientific life of our institute. We wish you an excellent start to the academic year.

Month in Review: IMSI Hosts IISE'26

A look back at the scientific events and activities that marked the life of our institute this month.

National Conference on Innovations in Industrial Safety and Environment (IISE'26)

On October 21, 2026, the Department of Industrial Safety and Environment at IMSI organized the National Conference on Innovations in Industrial Safety and Environment (IISE'26). The event was officially opened by Mr. Amani Ismail, Vice-Rector in charge of External Relations, Cooperation, Outreach, Communication, and Scientific Events.



The programme featured a keynote lecture on Industry 4.0 and sustainable industrial safety, highlighting the convergence of artificial intelligence and the Internet of Things (IoT) in support of the energy transition. The day also included oral scientific presentation sessions and poster sessions addressing key topics such as industrial risk prevention, environmental monitoring, sustainable resource management, and applications of artificial intelligence.

Startup & Basmat Mouhandis Expo

A highlight of the event was the exhibition of innovative projects developed by students and project holders. The Startup & Basmat Mouhandis Expo showcased technological solutions in industrial risk prevention and smart industrial safety.

This scientific gathering provided a valuable opportunity to foster academic exchange and to showcase the research and innovation driving our institute in the fields of industrial safety and the environment.

Month in Review: IMSI Hosts WICEM 2026

Another look at the scientific events and activities that marked the life of our institute this month.

Workshop on Instrumentation, Control, and Maintenance

IMSI hosted the WICEM 2026 Workshop on October 28, 2026, bringing together academic experts and industry professionals to discuss applied engineering solutions.

Organized at the IMSI - University of Oran 2 Mohamed Ben Ahmed, the workshop featured a series of specialized presentations bridging the gap between theoretical research and industrial application.



Program Highlights

The event was opened by the Institute Vice-Director at 9:00 AM. The scientific program included the following interventions:

- **Applications of Automation in Space Systems**
  A. Rabhi (Space Systems Development Center, Oran)
- **Instrumentation and Smart MRV for Agricultural Carbon Sinks**
  A. Benali (SONATRACH) and *M. Ait Mohamed* (INVA)
  Focus on the Decarbonatrach Scope 3 Program & ASSET Ecosystem
- **Management and Monitoring of Container Terminal Engines at the Port of Oran**
  K. Chergui (Port of Oran)
- **Industrial Engineering Applied to Healthcare Systems**
  K. Hachemi (Teacher-Researcher, IMSI)

Exhibition and Networking

Between the technical sessions, attendees had the opportunity to network during coffee breaks and explore the exhibition stands. The **Basmat Muhandis** and **MasterMind** clubs showcased their projects, highlighting the active role of student scientific clubs in the institute's innovation ecosystem.

The workshop concluded with a closing ceremony at 3:15 PM, marking a successful day of knowledge exchange and collaboration between academia and key national industry sectors.

Workshop WMMN'26: From Modeling to Modern Maintenance

The IMSI Exchange looks ahead to a new scientific event bridging applied mathematics, numerical simulation, and industrial maintenance.

The Institute of Maintenance and Industrial Safety (IMSI) at the University of Oran 2 Mohamed Ben Ahmed will host the **Workshop WMMN'26** on **Tuesday, November 3, 2026**.

Under the slogan *"Model • Simulate • Understand • Maintain,"* this event aims to explore the mathematical and numerical tools shaping the future of smarter and more sustainable maintenance. The workshop will bring together experts in applied mathematics, numerical modeling, and modern maintenance to share their research and practical applications.

Program Highlights

The day-long program will feature presentations from distinguished academics and researchers:

- **Prof. Habib Achache** (IMSI, University of Oran 2): *Maintenance and repair of aeronautical structures*
- **Prof. Seif-Eddine Bendaoui** (Relizane University): *Multi-scale optimization of FDM 3D printing parameters: Influence on mechanical properties, dimensional accuracy, and durability of functional parts*
- **Dr. Amina Boulekhrass** (IMSI, University of Oran 2): *Applied mathematics in maintenance*
- **Dr. Abdelkader Alioa** (Djillali Liabes University of Sidi Bel Abbes): *Numerical Study Traction*
- **Dr. Abdellah Feghoul**: *Role of the methods department in industrial maintenance*

Practical Workshop

A key highlight of the afternoon will be a practical session led by **Dr. Abd el Aziz Adjeloua** (IMSI): *Introduction to numerical simulation with Abaqus/CAE*. This hands-on workshop will al-

low participants to engage directly with industry-standard simulation software.

The IMSI Exchange encourages students, researchers, and staff interested in the intersection of mathematics, modeling, and maintenance to attend this enriching event.

Call for Papers: National Conference on Intelligent Production and Safety (NCIPS 2026)

The Maintenance in Instrumentation Department (MIT) of the Institute of Maintenance and Industrial Safety (IMSI), University of Oran 2 Mohamed Ben Ahmed, in collaboration with the LISIDD laboratory, invites colleagues, researchers, engineers, and doctoral students to participate in the National Conference on Intelligent Production & Safety (NCIPS 2026).

Event Information

Conference Date:	Nov. 26, 2026
Submission Deadline:	Nov. 15, 2026
Acceptance Notification:	Nov. 19-20, 2026
Mode	in person at IMSI

Conference Topics

- AI in Products & Manufacturing
- Process Optimization
- Intelligent Maintenance
- Industrial Safety & Risk

Practical Information

- **Registration Fees:** Free of charge,
- **Languages:** English, French,
- **Abstract Guidelines:** Maximum 250 words, with up to 5 keywords,
- **Paper Template:** Available on the conference website.

Submission and Contact

Submissions: https://ncips2026.sciencesconf.org
Contact: ncipsimsi@gmail.com

The IMSI Exchange encourages all interested members of our community to consider submitting their work and participating in this national event.

Call for Papers: National Conference on the Optimization of Industrial Systems (CNOSI'26)

The Department of Maintenance in Electromechanical Engineering at the Institute of Industrial Maintenance and Safety (IMSI), University of Oran 2 Mohamed Ben Ahmed, invites researchers, academics, and industry professionals to submit their work to the National Conference on the Optimization of Industrial Systems (CNOSI'26).

CNOSI'26 aims to provide a platform for exchanging ideas and presenting the latest research and industrial applications in the optimization, reliability, and safety of industrial systems.

Conference Topics

Submissions are invited in, but not limited to, the following thematic areas:

1. Predictive Maintenance, Digitalization & Industry 4.0

- Data-driven predictive maintenance
- Condition monitoring
- Prognostics and Health Management (PHM)
- Digitalization of maintenance processes and CMMS
- Artificial maintenance and machine learning
- Failure prediction and Remaining Useful Life (RUL) estimation
- Equipment protection, availability, and performance

2. Structural Integrity, Dependability & Safety

- Non-Destructive Testing (NDT) and advanced ultrasonics
- Structural integrity assessment and diagnostics
- Structural strengthening using composite materials
- Departs risk analysis and failure scenario assessment
- Industrial Asset Management (AMR) for risk management
- Dependability and major accident prevention

Submission Guidelines and Participation Formats

Authors are invited to submit their extended abstracts via the conference platform. All submissions will undergo peer review by the scientific committee.

Accepted contributions may be presented in the following formats:

- Keynote lectures
- Oral presentation sessions
- Poster presentation sessions
- Scientific club activities

Target Audience: Researchers, PhD students, students, startup founders, and industrial project leaders.
Participation Fee: Free of charge.

Key Dates

⚠ Dates subject to change.

Submission Deadline:	October 10, 2026
Acceptance Notification:	November 14, 2026
Registration Deadline:	November 21, 2026
Conference Date:	December 1st, 2026

Submission and Contact

Website: https://cnosi26.sciencesconf.org/

The IMSI Exchange encourages all eligible members of our community to consider submitting their research and participating in this national event.

Basma Gambit - Coming Soon!

Registrations for Basma Gambit will open soon. Stay tuned for more details!



I. SAIDI
3rd year industrial engineering student

Predictive Maintenance: Can We Predict When a Machine Will Break Down Before It Happens?

Have you ever wondered what would happen if a machine stopped working in the middle of a production line?

Imagine a factory where a machine worth millions suddenly stops because of a small component that costs very little to replace. How much would this failure cost the company in terms of money and time? This situation highlights the importance of anticipating failures before they occur. Today, thanks to AI, sensors, and data analysis, it is possible to detect these warning signs and predict potential failures before they happen. This approach is known as **predictive maintenance**. It relies on analyzing machine data to identify abnormal behavior, estimate the risk of failure, and help maintenance teams take action at the right time.

What Is Predictive Maintenance?

Sensors, captors, and AI are the major components involved in predictive maintenance. It is a modern technique that uses these elements to continuously monitor equipment conditions and accurately predict potential failures before they occur. Sensors collect essential operational data, such as vibration and temperature, which are then analyzed by machine learning algorithms to detect subtle abnormal changes that may indicate an emerging problem.

Predictive vs. Preventive Maintenance

Predictive maintenance differs from preventive maintenance, which relies on fixed schedules for carrying out maintenance activities. For example, in preventive maintenance, a component may be replaced even though it is still in good working condition. In contrast, predictive maintenance identifies the appropriate time for intervention based on the actual condition of the equipment.

Practical example: An industrial motor can be monitored by analyzing its vibrations to detect signs of bearing wear before the problem becomes more serious. This helps reduce unexpected downtime and lower maintenance costs.

How Does Artificial Intelligence Work?

To understand the core mechanisms of AI in this domain, we must examine the major steps involved in an AI setup and implementation process:

- **Data Collection:** Sensors are installed on the machines to measure vibration, temperature, pressure, and other parameters, depending on the type of equipment.
- **Data Analysis:** The system analyzes the collected data and compares it with normal operating values. If the measurements differ from the usual values, this may indicate a potential problem.

- **Machine Learning:** The artificial intelligence model is trained using historical data, including data from normal operating conditions and periods when failures occurred. The model learns patterns associated with failures and uses them to estimate the likelihood of a new failure.
- **Warning and Diagnosis:** When the system detects abnormal indicators, it sends an alert to the maintenance team, identifying the potential type of failure.



Vibration Analysis

Vibration analysis is a specific technique used within predictive maintenance to monitor machines, such as motors and pumps, and detect faults before they occur. Sensors measure machine vibrations, and the data is analyzed to identify problems such as imbalance, misalignment, and bearing wear.

Benefits:

- Early fault detection
- Continuous machine monitoring
- Reduced maintenance costs and unexpected downtime

A Real-Life Example: Sachsenmilch

While researching AI applications in industry, I came across an interesting example from Sachsenmilch, a dairy factory in Germany. The factory used Siemens' AI-powered system, *Senseye Predictive Maintenance*, to monitor its equipment and detect potential failures. The system identified a pump nearing the end of its service life, allowing the factory to plan its replacement before an unexpected breakdown occurred.

I find this example fascinating because it shows how AI can help factories save money, reduce downtime, and make smarter maintenance decisions. Beyond this example, predictive maintenance helps industries detect problems early, reduce costs, prevent unexpected breakdowns, and improve safety. I believe that AI is changing the future of industry, making machines smarter and factories more efficient. The best way to fix a problem is to prevent it before it happens.

References

- IBM. *What Is Predictive Maintenance?* https://www.ibm.com/think/topics/predictive-maintenance
- Siemens. *Sachsenmilch: AI-Based Predictive Maintenance.* https://www.siemens.com/en-us/company/insights/sachsenmilch-dairy-predictive-maintenance/

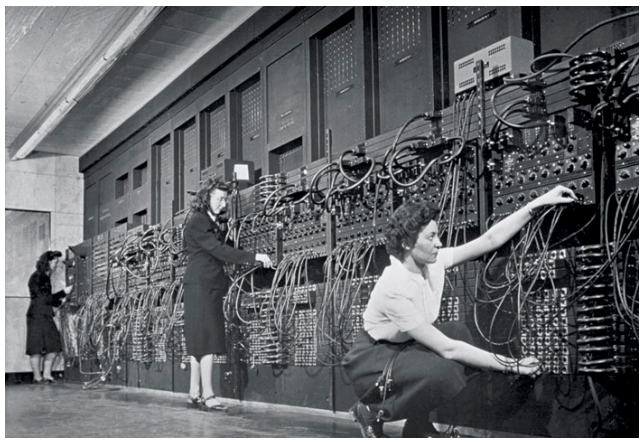


Z. BENDANI
3rd year engineering student

Industry 3.0: From the ENIAC Behemoth to Toyota's Lean Machine

Perhaps, my friend, you are reading this article right now from a computer screen. If so, let me tell you that John Mauchly and J. Presper Eckert were the ones who invented the first programmable electronic computer, the ENIAC. This took place during World War II, when the US Military desperately needed to compute artillery firing tables-calculations that previously took human "computers" days to complete.

Their computer was a 27-ton behemoth occupying an entire room! Yet over the decades, computers shrank until they weighed less than a single kilogram. What was the secret behind this dramatic shift in size?

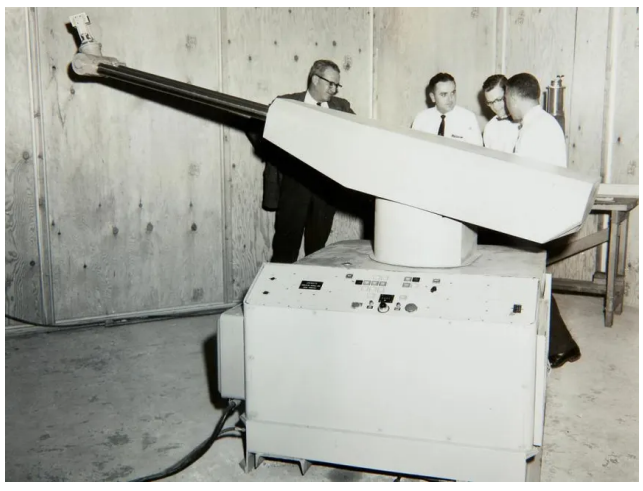


ENIAC

The secret lies in a tiny component called the Transistor-arguably the most important invention of the 20th century. It was created at Bell Labs by the legendary trio: John Bardeen, Walter Brattain, and William Shockley.

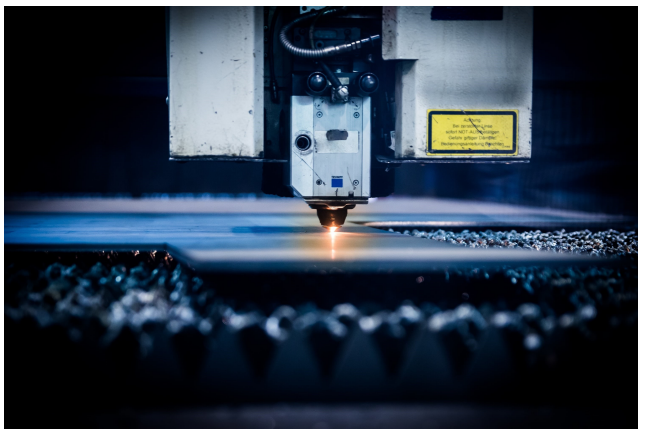
The Era of Robotics and Automation

In 1961, the world's first industrial robot, Unimate (invented by George Devol and Joseph Engelberger), officially joined the assembly line at a General Motors plant in New Jersey.



UNIMATE

Equally fundamental to this era was the PLC (Programmable Logic Controller): an industrial computer engineered to withstand harsh factory environments (heat, dust, vibration) and automate machinery without manual wiring. In 1968, American engineer Dick Morley invented the first PLC, the Modicon 084. This eliminated the nightmare of endless relay control wiring and paved the way for advanced CNC (Computer Numerical Control) machinery.



Japanese Philosophy and Integrated Manufacturing

While automation delivered staggering efficiency, Western factories soon faced a new hurdle: massive inventory accumulation.

Entering the scene, Toyota engineers Taiichi Ohno and Shigeo Shingo re-architected production concepts into the Toyota Production System (TPS), globally known today as Lean Production.

Simultaneously, software capabilities surged with CAD/CAM (Computer-Aided Design and Manufacturing) and ERP (Enterprise Resource Planning) systems.

The pinnacle wave of Industry 3.0 was CIM (Computer-Integrated Manufacturing), where the entire factory operated as a single living organism: ERP served as the brain, local networks (LAN/Fieldbus) as the nervous system, and PLCs, CNCs, and Robots as the working hands.

We have only touched on the surface of this fascinating era, my friend, and I encourage you to explore each milestone further. See you in the next article, where we step through the gates of Industry 4.0!





I. DROUICHE
3rd year industrial engineering student

Being Afraid to Lose a Person Can Be What Makes That Person Leave

> *May the petals teach me the art of letting go.*
>
> – XAN OKU

The concept of love has been addressed by scientists and philosophers for centuries. By "love," it does not necessarily mean romantic love; for example, the relationship between us and our kids, siblings, friends, and neighbors is also an act of love. The Greek philosopher Plato believed that "love makes

us whole again"-the idea that relationships are meant to help a person become complete and assemble the pieces of his true self. This idea has been supported by a number of philosophers, such as Russell and de Beauvoir, who saw that we love in order to satisfy our physical and psychological desires and that, because of our cold, cruel world, we tend to shelter ourselves in isolation, fearing the world and trying to protect ourselves. Love's delight, intimacy, and warmth help us overcome our fear of the world, escape our lonely shells, and engage more abundantly in life. It then infuses our lives with meaning and lets us reach beyond ourselves.

But we may encounter cases where we love and care so much about another person that we forget our duties and our life just for that person-and yet we end up breaking up with him and leaving ourselves completely destroyed. Or perhaps the opposite happens: we are the one being loved and cared about, but eventually we end up leaving that person. In everyday language, this pattern is sometimes called "anxious love." More accurately, it involves anxious attachment and emotional dependency.

The attachment is the feeling that life relies only on one person, that he is the only light in a dark world, and that we must keep this light attached to us or we will wander in darkness. Because of this thinking, we may find ourselves giving everything we have (like time, money, and emotions) to that person in order to receive a small compliment, attention, or just a smile. We may worry about the tiniest thing that might cause him to leave, or we may become so suspicious about whether he still loves us that we ask questions such as: "Why hasn't he responded yet?" "Why is he ignoring me, even if he isn't?" "Why can't he see my endless efforts and sacrifices for him?" Such questions are constantly asked by the anxiously attached person, who may then use them as an excuse to suspect the other person's love and potentially to break the relationship.

In psychology, attachment is not a "level" of love. Attachment is a bond or security system: it describes how we seek closeness, safety, and reassurance from important people. Attachment styles are patterns of relating, not fixed personality types. A secure attachment style means a person is generally comfortable with closeness and independence. Insecure patterns include fear and anxiety. One insecure pattern is avoidant attachment. In this pattern, a person becomes attached to someone but, when things get more intense, he withdraws because, well, it got too intense, as he believes. The thing is that this person is afraid of love, and pulling away is just his way of dealing with this fear. It is not that he doesn't want love; he wants it and may crave it, but at the same time he is terrified of it. Why? Because it did not go well for him in the past, and mostly in childhood, where love was seen as a threat.

If you grow up in an environment where love wasn't possible, where relationships between caregivers and children were not safe, you may learn to classify love as a risk. You then develop a defense mechanism against this risk, and by the time you reach adulthood you find yourself unconsciously defending yourself from this risk-which is love-and therefore you may find yourself sabotaging true relationships. That is what can happen to the avoidantly attached person. What we need to understand is that in love we don't only look for what makes us happy; we also look for what feels more familiar. So for someone whose most familiar feeling is a sense of not being loved properly, a sense of someone threatening us, or a sense of someone who may abandon us, that can feel like his true home. He cannot get over it unless he gets honest with himself. This is a tendency, not a destiny.

The second insecure pattern is anxious attachment. Instead of running from love, as described above, the anxiously attached person puts his best effort in just so he can receive love from a specific person. He becomes like a candle that burns itself in order to give light to its environment. A major reason for such attachment is that the attacher did not receive much consistent love in childhood from his parents or family members and may have had toxic relationships that used him in the past. This person then loses confidence and feels inadequate, and so a hole is created in his self. This deficiency makes him crave something to fill that hole, and when the relationship reaches that state, those emotions and feelings that he wished for from childhood and was deprived of start pouring in. They automatically fill that hole-and those feelings become addictive.

At the beginning it looks charming: a person who seeks love finally gets someone who cares about him, and so they get along and spend a lot of time together. But as time passes, it becomes consuming for both people in the relationship. For instance, the addiction to emotions that the attacher has will keep going on and on, and so he will constantly want to receive them more and more. Here it gets more intense, because for the other person this is fatiguing. You see, relationships are built on the principle of giving and taking, but here he is always constantly giving emotions and receiving little, even if the attacher gives his time, acts of service and gifts. But all of this is just so he can receive more emotional reassurance, so in this case those sacrifices do the opposite of the role they should have.

It is much like holding a bird: instead of holding it gently, the attacher strangles the bird, thinking it is better so it can't leave; but the moment it slips away, the bird will immediately fly away. What makes it worse is that when the attacher notices the other person pulling away, he will try more and more to stick to him. He will sacrifice more and try to be more emotional so he can stay, not realizing that this action itself is what pushes that person away. And because he depends on him so much, life feels empty without him.

You see, what the attacher doesn't get is that he is not attached to the person itself but rather to those feelings that he receives, because he lost faith in himself long ago. That attention he gets from those feelings is what makes him feel his own value and thus love himself; he is seeing himself through that person. That is why the person appears to be everything in his life. By letting go of those emotions, he will lose faith in himself as he was before. It is like giving up on a piece of yourself that you think you'll die without. Eventually the relationship will reach the breaking point, and the attacher will either try more to get the person back (which will push the other person further away) or will hate the person and desperately look for another source in which he sees himself. Either way, he will be left in fear and sadness.

To heal from unhealthy attachment, you have to look at your past and reconnect with the true self that has been buried over time. You have to recognize the moments and experiences that made you feel inadequate and caused you to lose confidence. You have to accept your personality, with its flaws, and then do what you can to shape it for the better. You have to fill that hole yourself instead of relying on others to fill it, and above all, you need to love yourself. You do not have to sacrifice yourself for others, and they should not do the same for you. You must learn how to love and how to maintain a healthy relationship. Love is not built on fear and anxiety but on confidence and peace. It is not about who burns himself out the most for the other, but about supporting each other.

There is a strong chance that you have experienced unhealthy attachment in one form or another-whether toward a person, an object, or an idea. If so, it is a sign that you need to examine the reasons behind it and be honest with yourself, as Ram Dass said: "A feeling of aversion or attachment toward something is your clue that there's work to be done."

Science is elegant.



F. KEDDAR GARCIA
3rd year industrial engineering student

SOME OF THE MOST IMPORTANT THINGS IN OUR LIVES ARE THE ONES WE TAKE FOR GRANTED

There are things that have accompanied humanity since the dawn of history. Their presence has become so familiar that we rarely stop to notice them. We see them every day until they become part of the background, as if they have always been there... and always will be. One of them is a celestial body that has shared our sky for billions of years.

The Moon has orbited Earth for about $4.5\,\text{billion}$ years, held in its orbit by gravity. Over time, its presence has become so ordinary that most people never stop to think about its true role or how deeply it influences our planet. Yet some of the most important things in our lives are the ones we take for granted.

What seems like a permanent part of the night sky may actually be one of the invisible pillars that help keep Earth in balance. But what if the Moon suddenly disappeared? Would we simply lose the bright object that lights up our nights, or would our planet face consequences far greater than we ever imagined?

To better understand what would happen if the Moon disappeared, let us follow the event step by step and examine how our planet would respond over time.

AFTER ABOUT 1.3 SECONDS...

The first consequence of the Moon's disappearance would not be felt by humans. In fact, we would most likely notice nothing at all at first. The human body is dominated by Earth's gravity, which produces an acceleration of $9.81\,\text{m/s}^2$, whereas the Moon's gravitational acceleration at Earth's surface is only about $3.3 \times 10^{-5}\,\text{m/s}^2$, roughly $0.000\,34\,\%$ of Earth's gravity. This difference is far too small for us to feel directly.

However, this is the moment when the first real consequence begins. The gravitational interactions that have shaped the Earth-Moon system for billions of years would suddenly lose one of their most important components. With a mass of $7.35 \times 10^{22}\,\text{kg}$,

the Moon continuously exerts a gravitational pull on Earth.

According to Newton's law of universal gravitation, any two objects with mass attract each other, regardless of how weak that force may be. Even so, Earth would not drift out of its orbit around the Sun or wander through space, as is often imagined. The Sun's gravity overwhelmingly dominates Earth's orbit. The Moon's role is not to keep Earth orbiting the Sun, but to subtly influence Earth's motion and contribute to the long-term dynamics of our planet.

After One Hour...

One hour after the Moon disappears, about $3600\,\text{seconds}$ have passed since Earth lost its gravitational influence. Yet we should not expect dramatic events to unfold immediately. Nature does not change everywhere at once, and the oceans are no exception.

For billions of years, Earth's oceans have been shaped by the combined gravitational effects of the Moon and the Sun. Their motion cannot suddenly stop or reverse. Instead, the oceans would begin a gradual adjustment, responding to a new reality as water redistributed under the remaining forces acting upon it.

To understand what would happen over the coming hours, we must first understand why the Moon has such a strong influence on the oceans. At first glance, this may seem confusing. If Earth's gravity is far stronger than the Moon's, why isn't Earth itself responsible for the tides?

The answer lies in the fact that the Moon does not pull equally on every part of our planet. The side of Earth closest to the Moon experiences a slightly stronger gravitational pull than Earth's center, while the far side experiences a slightly weaker pull. This tiny difference in gravitational force is known as the *tidal force*, and it is what creates the ocean tides.

Earth's own gravity, despite being much stronger, pulls the oceans and the planet itself almost equally. Because there is little difference in its pull across Earth's surface, it cannot generate the same tidal effect.

But what happens when the force that has driven this cycle for billions of years suddenly disappears? Will the oceans stop their endless rhythm of rising and falling, or will they continue to move in a different way?

As lunar tides weaken and the oceans adjust, tidal movements would change, potentially affecting the movement of water and nutrients in coastal ecosystems. These changes could influence feeding and reproduction in some marine species. Although the effects would vary from one ecosystem to another, they would not represent the Moon's only important legacy. Another consequence could unfold far more slowly, over thousands or even millions of years, as changes in Earth's long-term dynamics begin to influence our planet.

After One Month...

One month after the Moon's disappearance, Earth would still be adjusting to a new reality. With the Sun now acting as the only remaining source of astronomical tidal forces, the oceans would continue responding to the changed tidal pattern.

This does not mean that the water suddenly stopped moving when the Moon vanished. Instead, the oceans would continue moving because of inertia, a fundamental principle of Newton's first law of motion, which states that an object remains at rest or continues moving at a constant velocity unless acted upon by a net external force.

Before the Moon disappeared, Earth's tides were produced by the combined gravitational effects of the Moon and the Sun, with the Moon providing the stronger contribution. Now, the lunar component would be gone, leaving only the Sun's tidal influence. The oceans would gradually respond to this change, although the resulting patterns would differ from one region to another.

But the consequences could extend far beyond the oceans themselves. For millions of years, countless coastal organisms have evolved in harmony with the rhythm of the tides. To us, tides may seem like nothing more than the regular rise and fall of sea level. For many marine species, however, they serve as a biological clock that regulates feeding, movement, and reproduction.

Take mussels, for example. These shellfish spend their lives attached to rocks along the shoreline. As the tide rises, seawater brings oxygen and tiny food particles that they filter from the water. When the tide retreats, they close their shells to conserve moisture and wait for the next high tide.

With changing tidal patterns, some shorelines might be covered by seawater less frequently or for shorter periods, affecting the conditions in which these animals live and feed. The impact would not end there. Many marine organisms time the release of their eggs or reproductive cells according to environmental cues, including tidal cycles, allowing moving water to carry them over greater distances.

Changes in these rhythms could affect some species' reproductive success, while other organisms might adapt or respond differently. The consequences would depend on the species and the environment in which they live.

The Moon's importance lies partly in the fact that it has carried out its role silently, without demanding our attention. We become so accustomed to its presence that we stop asking why it matters in the first place. Yet the things we take for granted are often the very things that help make our world possible.

Thousands of Years Later...

Thousands of years after the Moon's disappearance, its greatest impact might no longer be seen only in the oceans or along the coastline, but in the long-term behavior of Earth itself.

This leads to a fascinating question that few people ever think to ask: what has the Moon been doing for Earth over the past $4.5\,\text{billion}$ years without us even noticing?

To answer that question, we first need to understand Earth's axis.

Earth rotates once every $24\,\text{hours}$ around an imaginary line that passes through the North and South Poles. However, this axis is not perfectly upright. It is tilted by about $23.44°$ relative to a line perpendicular to Earth's orbital plane around the Sun.

Although this tilt may seem like a small detail, it is one of the main reasons Earth experiences seasons as we know them. Because of this tilt, different parts of Earth receive different amounts of sunlight throughout the year. When one hemisphere tilts toward the Sun, it receives more direct sunlight and experiences summer, while the opposite hemisphere tilts away, bringing winter.

About six months later, the situation reverses, creating the cycle of the seasons.

But if Earth's axis is tilted, why has its angle remained within a relatively narrow range for long periods? Why does it change under the gravitational influence of the Sun and the other planets?

The answer partly lies in Earth's shape. Our planet is not a perfect sphere. Because it rotates, it bulges slightly around the equator and is slightly flattened at the poles. Both the Moon and the Sun exert gravitational forces on this equatorial bulge, producing what scientists call *gravitational torque*-a turning effect that can change the orientation of a rotating object rather than simply pulling on it.

Although the Sun is vastly more massive than the Moon, the Moon's much closer distance gives it a crucial role in stabilizing Earth's axial tilt. As a result, Earth's tilt has remained within a relatively narrow range over long periods of time. Today, it varies between about $22.1°$ and $24.5°$ over a cycle of roughly $41\,000\,\text{years}$, and the Moon plays a major role in limiting these variations.

If the Moon were to disappear, Earth's axis would not suddenly change overnight. No immediate, dramatic difference in the seasons would occur within days or years. However, over much longer periods, the stabilizing influence of the Moon would be gone, allowing Earth's axial tilt to undergo different variations under the gravitational influence of the Sun and other planets.

Scientists have studied hypothetical Earths without a Moon and found that their axial tilts can vary more than Earth's does today, although the extent of those changes depends on the conditions of the system. Therefore, we cannot predict one exact future for Earth's axis.

If its tilt changed significantly, the consequences could extend far beyond the axis itself. The intensity and distribution of the seasons could change dramatically. Some regions might experience hotter summers and colder winters, while others could develop very different seasonal patterns.

Over geological timescales, changes in the distribution of sunlight could influence rainfall patterns, the extent of deserts and ice sheets, and the development of entire ecosystems. Earth's climate would not necessarily become uninhabitable, but it could become very different from the climate we know today.

Conclusion

In the end, the Moon was never just a beautiful object in the night sky or the force behind the tides. For billions of years, it has influenced Earth's oceans and helped stabilize the long-term behavior of our planet's axis, contributing to the conditions under which life has evolved.

So, the next time we look up at the sky, we may no longer see the Moon as just a bright companion of the night. Instead, we may see it as a silent guardian of Earth's delicate balance-a reminder that the greatest forces in nature are not always the loudest, but often the ones working quietly in the background, sustaining our world every single day.